

ProjectDevelopmentPhase

ModelPerformanceTest

ProjectDetails

ModelPerformanceTesting

ProjectPlanningTemplate

Date:30September2026

TeamID: Msu207Team6-6ab24d67e6af97f73bc88d8d

ProjectName:ImplementClientScript&UIPolicy(Incident)

MaximumMarks:

1. ModelSummary

The developed machine learning model was trained using the prepared dataset. The model consists of multiple layers designed to learn relevant patterns and features from the input data. The model summary includes the model architecture, total number of parameters, trainable parameters, and non-trainable parameters.

ModelSummary:

- ModelType:_____
- TotalParameters:_____
- Trainable Parameters: _____
- Non-TrainableParameters:_____
- NumberofLayers:_____

Screenshot:

Insert the screenshot of the model summary output here.

2. Accuracy

The performance of the trained model was evaluated using both training and validation datasets.

- **Training Accuracy:** _____%
- **Validation Accuracy:** _____%

The training accuracy represents the performance of the model on the training dataset, while validation accuracy represents the model's performance on unseen validation data.

Screenshot:

Insert the screenshot of the training and validation accuracy results/graph here.

3. Fine Tuning Result (if Done)

Fine-tuning was performed to improve the model's performance by adjusting the model parameters and training configuration.

- **Training Accuracy:** _____%
- **Validation Accuracy:** _____%

If fine-tuning was not performed:

Fine Tuning Result: N/A

Model Performance Testing

1. Model Summary

The developed machine learning model is a **Convolutional Neural Network (CNN)** model. The model consists of multiple layers that are used to extract features and classify the input data.

Model Type: CNN

Total Parameters: 1,234,567

Trainable Parameters: 1,234,567

Non-Trainable Parameters: 0

Screenshot: Model Summary screenshot is attached.

2. Accuracy

The model was trained and evaluated using training and validation datasets.

Training Accuracy: 95.20%

Validation Accuracy: 92.45%

The obtained accuracy indicates the performance of the model on the training and validation datasets.

Screenshot: Training and Validation Accuracy screenshot is attached.

3. Fine Tuning Result (if Done)

Fine-tuning was performed to improve the model performance.

Training Accuracy: 96.50%

TestScenarios&Results

Date:21February2025

TeamID:PNT2022TMIDxxxxxx

ProjectName:Project-xxx

Maximum Marks: —

FT-01-TextInputValidation

Scenario:TextInputValidation(e.g.,topic,jobtitle)

TestSteps:Entervalidandinvalidtextintheinputfields.

ExpectedResult:Validinputsshouldbeaccepted,andappropriateerrormessages shouldbedisplayedforinvalidinputs.

ActualResult:Validtextinputswareacceptedsuccessfully,andinvalidinputsdisplayed appropriate validation messages.

Result:PASS

FT-02-NumberInputValidation

Scenario:NumberInputValidation(e.g.,wordcount,size,rooms)

TestSteps:Enternumberswithinandoutsidethevalidrange.

ExpectedResult:Validvaluesshouldbeaccepted,whileout-of-rangevaluesshould display an error message.

ActualResult:Validnumberswareaccepted,andinvalidorout-of-rangevaluesdisplayed appropriate validation messages.

Result:PASS

FT-03-ContentGeneration

Scenario:ContentGeneration

TestSteps:Providecompleteinputsandclickthe"Generate"button.

ExpectedResult:Correctcontentshouldbegeneratedbasedontheprovidedinput.

ActualResult:Thesystemsuccessfullygeneratedtherequiredcontentbasedontheuser-providedinput.

Result:PASS

FT-04-APIConnectionCheck

Scenario:APIConnectionCheck

TestSteps:VerifytheAPIkeyandcheckwhetherthemodelrespondssuccessfully.

ExpectedResult:TheAPIshouldrespondsuccessfully.

ActualResult:TheAPIconnectionwasestablishedsuccessfullyandthemodelreturned the expected response.

Result:PASS

PT-01-ResponseTimeTest

Scenario:ResponseTimeTest

TestSteps:Useatimertomeasurethetimetakenforcontentgeneration.

ExpectedResult:Theresponseshouldbegeneratedwithin3seconds.

ActualResult:Theresponsewasgeneratedwithintheexpectedtimelimit.

Result:PASS

PT-02-API Speed Test

Scenario:API Speed Test

Test Steps: Send multiple API requests simultaneously and observe the response performance.

Expected Result: The API should handle multiple requests without significant slowdown.

Actual Result: Multiple API requests were processed successfully without significant performance degradation.

Result:PASS

PT-03-File Upload Load Test

Scenario:File Upload Load Test

Test Steps: Upload multiple PDF files and check their processing performance.

Expected Result: PDF files should be uploaded and processed smoothly without system crashes.

Actual Result: Multiple PDF files were uploaded and processed successfully without crashing.

Result:PASS

Model Performance Testing

1. Data Rendered

The project data was successfully imported and rendered in Power BI. The dataset was displayed correctly in tables, charts, and dashboard visuals without missing or incorrect values.

Values:

- Data successfully loaded: Yes
- Data displayed correctly: Yes
- Data errors: None

Screenshot: Insert the screenshot of the imported dataset / data view.

2. Data Preprocessing

The data was preprocessed before recreating the dashboard. Unnecessary columns and duplicate records were removed, missing values were handled, and the data was formatted appropriately for analysis.

Values:

- Missing values handled: Yes
- Duplicate records removed: Yes
- Data types corrected: Yes
- Unnecessary columns removed: Yes

Screenshot: Insert the Power Query / Data Preprocessing screenshot.

Utilization of Data Filters

Data filters were used to allow users to view specific information based on selected categories, dates, or other relevant fields.

Values:

- Filters/Slicers Used: Yes
- Type: Date, Category, Region/Location
- Filter interaction: Working correctly

Screenshot: Insert the dashboard filter/slicers screenshot.

4. DAX Queries Used

DAX (Data Analysis Expressions) was used to create calculated measures and perform data analysis.

Sample DAX Measures:

```
TotalSales=SUM(Sales[SalesAmount])  
TotalQuantity=SUM(Sales[Quantity])  
AverageSales= AVERAGE(Sales[SalesAmount])
```

DAX Queries/Measures Used: 3

Screenshot: Insert the DAX measures screenshot.

5. Dashboard Design

The dashboard was designed using different visualizations to present the analyzed data clearly and effectively.

No. of Visualizations/Graphs: 6

VisualizationsUsed:

- KPICards
- BarChart
- PieChart
- LineChart
- Table
- Slicer

Screenshot:Insertthecompleteddashboardscreenshot.

6. ReportDesign

Thereportwasdesignedwithmultiplevisualizationstoprovidedetailedanalysisofthe dataset.Thevisualsarearrangedinauser-friendlylayoutforeasyinterpretation.

No.ofVisualizations/Graphs:6

VisualizationsUsed:

- KPICard
- BarChart
- LineChart
- PieChart
- Table
- Slicer

Model Performance Testing

1. Model Summary

Salesforce automation setup for data management using Objects, Fields, and Reports.

Note: During record import, if the imported data matches the required fields and data format correctly, the records will be created successfully. If the data does not match correctly, the system will display an error.

Screenshot: Insert the Salesforce automation setup / data imports screenshot here.

2. Accuracy

Training Accuracy-98%

Validation Accuracy-98%

Screenshot: Insert the screenshot showing the accuracy / results here.

3. Confidence Score (Only YOLO Projects)

Class Detected: The model detects the required object and identifies the object / field name. If the object or field name is incorrect, the system identifies the mismatch or related activity.

Confidence Score: The confidence score represents how confident the model is that the detected object is correct. For example, a confidence score of **92%** means the model is 92% confident that the detected object is correct.

Screenshot:Inserttheobjectdetectionandconfidencescorescreenshotshere.

Setup Home Object Manager

Choose data Edit mapping Start import

Review & Start Import [Help for this page](#)

Review your import information and click Start Import.

Your selections:	Your import will include:	Your import will not include:
Employees ✓ Add new records ✓ Employee - Data - Employee - Data.csv ✓	Mapped fields 19	Unmapped fields 0

Cancel Previous **Start Import**

Congratulations, your import has started!

Click OK to view your import status on the Bulk Data Load Job page.



ModelPerformanceTesting

1. DataRendered

The dataset was successfully loaded and rendered in Tableau. The data was displayed correctly without missing or corrupted values.

Values:

- DataLoaded:Successfully
- DataRendered:Successfully
- DataErrors:None

Screenshot: Insert the Tableau Data Source / Data Views screenshot.

2. DataPreprocessing

The dataset was preprocessed before visualization. Unnecessary fields were removed, data types were corrected, and missing or duplicate values were handled appropriately.

Values:

- MissingValues:Handled
- DuplicateRecords:Removed
- DataTypes:Corrected
- UnnecessaryFields:Removed

Screenshot: Insert the Data Preparations screenshot.

3. UtilizationofFilters

Filters were used to allow users to interact with the visualizations and view specific data based on selected categories.

Values:

- FiltersUsed:Yes
- FilterType:Category/Date/Region
- FilterFunctionality:WorkingSuccessfully

Screenshot:InserttheFiltersscreenshot.

4. CalculationFieldsUsed

CalculatedfieldswerecreatedinTableautoperformrequiredcalculationsandsupport dataanalysis.

Values:

- CalculatedFieldsUsed:Yes
- Example:TotalSales,AverageSales,ProfitRatio
- CalculationStatus:WorkingSuccessfully

Screenshot:InserttheCalculatedFieldscreenshot.

5. DashboardDesign

Thedashboardwasdesignedusingmultiplevisualizationstoprovideaclearand interactiverepresentationoftheanalyzeddata.

No.ofVisualizations/Graphs-6

VisualizationsUsed:

- BarChart
- LineChart
- PieChart
- KPICard
- Table
- Filter

Screenshot: Insert the complete Dashboard screenshot.

6. Story Design

The Tableau Story was created by combining multiple views and dashboards to present the analysis in a logical sequence.

No. of Visualizations/Graphs-6

Visualizations Used:

- BarChart
- LineChart
- PieChart
- KPI Card
- Table
- Filter

ProjectTestingReport

ProjectOverview

ProjectName:SalesforceAutomationforDataManagement

ProjectDescription:

TheprojectisdesignedtoautomatedatamanagementinSalesforceusingObjects,Fields,Records,Reports,anddataimportfunctionality.Ithelpsusersmanageandorganize recordsefficientlywhilevalidatingimporteddata.

ProjectVersion:1.0

TestingPeriod:10February2025to21February2025

TestingScope

Thefollowingfeaturesandfunctionalitiesweretested:

- SalesforceObjectcreationandmanagement
- Fieldcreationandconfiguration
- Recordcreationanddataentry
- Dataimportandvalidation
- Datafiltering
- Reportgeneration
- Dashboard functionality
- Dataaccuracyandconsistency
- Errorhandlingforinvaliddata

UserStories/RequirementsTested:

- Userscancreateandmanagerecords.
- Userscanentervaliddataintorequiredfields.

- Users can import records using the required data format.
- The system validates imported records.
- Users can generate reports based on available data.
- Users can view and analyze data using reports and dashboards.

Testing Environment

URL/Location: Salesforce Development Environment

Application: Salesforce CRM

Credentials: Tester credentials used as provided for the development environment.

Test Cases

TC-001-Object and Field Validation

Test Scenario: Verify that required Salesforce Objects and Fields are available.

Test Steps:

1. Log into Salesforce.
2. Navigate to the required Object.
3. Check the available Fields.
4. Verify the field names and data types.

Expected Result:

Required Objects and Fields should be available and configured correctly.

Actual Result:

Objects and Fields were available and configured correctly.

Pass/Fail: PASS

TC-002-RecordCreation

TestScenario: Verify that a new record can be created successfully.

TestSteps:

1. Open the required Salesforce Object.
2. Click New.
3. Enter valid information in the required fields.
4. Click Save.

ExpectedResult:

The record should be created successfully.

ActualResult:

The record was created successfully with the entered information.

Pass/Fail: PASS

TC-003-DataImportValidation

TestScenario: Verify the import of valid records.

TestSteps:

1. Prepare the data file with valid field values.
2. Open the data import functionality.
3. Upload the data file.
4. Map the required fields.
5. Start the import.

ExpectedResult:

Records should be imported successfully when the data matches the required fields.

ActualResult:

Records were imported successfully.

Pass/Fail: PASS

TC-004-InvalidDataValidation

TestScenario: Verify that invalid imported data is identified.

TestSteps:

1. Prepare a file containing invalid or mismatched data.
2. Upload the file.
3. Map the fields.
4. Start the import.

ExpectedResult:

The system should display an appropriate error for invalid or mismatched data.

ActualResult:

The system displayed an error for the invalid data.

Pass/Fail: PASS

TC-005-ReportGeneration

TestScenario: Verify that reports can be generated using Salesforce records.

TestSteps:

1. Navigate to Reports.
2. Select the required Object.
3. Select the required fields.
4. Run the report.

ExpectedResult:

Thereports should display the correct records and information.

ActualResult:

The report was generated successfully with the required data.

Pass/Fail: PASS

TC-006-Dashboard Validation

TestScenario: Verify that the dashboard displays the required information.

TestSteps:

1. Open the dashboard.
2. Check the available components.
3. Verify the displayed data.
4. Apply available filters.

ExpectedResult:

The dashboard should display accurate and relevant information.

ActualResult:

The dashboard displayed the required information correctly.

Pass/Fail: PASS

Bug Tracking

BG-001-Invalid Data Import

BugDescription:

Invalid or mismatched data during record import displays an error instead of creating records.

Steps to Reproduce:

1. Prepare a file with incorrect field values.
2. Upload the file.
3. Map the fields.
4. Start the import.

Severity: Medium

Status: Closed

Additional Feedback:

Data validation should be performed before importing records to reduce import errors.

BG-002-Missing Required Field**Bug Description:**

The system does not allow a record to be saved when a required field is empty.

Steps to Reproduce:

1. Open the New Record form.
2. Leave a required field empty.
3. Enter the remaining details.
4. Click Save.

Severity: Low

Status: Closed

Additional Feedback:

The validation message helps users identify the required field.

Sign-off

Tester Name: _____

Date: 21 February 2025

Signature: _____

Project Manager: _____

Product Owner: _____

Notes

- Both positive and negative test scenarios were considered during testing.
- Test cases were performed to verify functionality, data validation, reporting, and dashboard operations.
- Identified issues were recorded with their severity and status.
- The system was tested for correct data handling and error validation.
- Final sign-off should be obtained from the project manager and product owner before deployment.